

REV. 00



PLAN REBUILD CENTER

CYLINDER HEAD

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

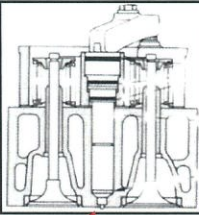
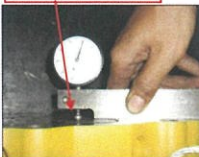
MACHINE MODEL

WO NO:

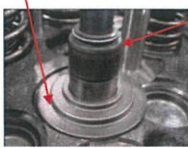
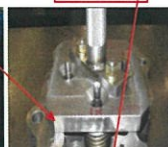
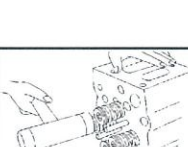
SUB ASS'Y SECTION

ASSEMBLY CHECK SHEET CYLINDER HEAD ENGINE ALL SERIES	WO NO.		PART NAME	CYL. HEAD ASS'Y	DOCUMENT NO.	RC/QR04/EG/03/03
	ENGINE MODEL	SA6D140-2	PART NUMBER		PUBLIC DATE	13/10/2009
	ENGINE S/N		START DATE /TIME		REVISION NO.	00
	MACHINE MODEL	GD 825-2	FINISH DATE / TIME		PAGE	2 of 2

ASSEMBLY BY						All dimensions in millimetres
						All torques in Kilogram metres

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS & COMMENT)
					MEC	QA	SPV	
1	CYL.HEAD ASSEMBLY	Head Assy Threads Cyl.Heads	- Head Assy must be free from crack, dents, pitting, rust and worn - Make sure all threads are not damage - Make sure top surface and bottom surface of the Cylinder heads are good conditions - Make sure the Cylinder Heads are clean					
2	VALVE GUIDES	Bores	- The Valve Guides be renew after measured. - Make sure the Valve Guide bores are clean when reinstall of the Valve Guides • Before inserting the valve guide, coat the outer dimension of the valve guide with small quantity of engine oil (SAE 15W-40)	GO and NO GO Hyd. Press				  Insert and press valve guides
3	INSERT VALVE	Insert Valve Bores Valve Sinking	- Make sure the dimension of the Insert Valve bores - Check that there is no dust, crack, corrosive and dirt - Install the Insert Valve be carefully • Diameter between Insert Valve bore and Insert Valve must interference see measurement check sheet • When install Insert Valves must straight to hole position - Check Valve sinking/Protrusion and refer to Standard (Shop Manual or measurement Check Sheet) • If sinking or Protrusion out of standard, cutting Insert or Valve and recheck sinking or Protrusion it must standard condition.	Special Tool Grinding Valve machine or Cutting Insert machine				 Install insert valve  Check valve sinking 
4	NOZZLE SLEEVE	Nozzle Sleeve Condition	- The Nozzle Sleeves must be renew - Check the O-ring grooves and the Nozzle Sleeve bores are clean when reinstall of the Nozzle Sleeves - Make sure the Nozzle Sleeves are in good position • When install and cutting the Nozzle Sleeves must use the Special Tool and refer to replacing Nozzle Holder Sleeve in Shop Manual (see replacing nozzle holder sleeve Job Instruction). • Check and measure protrusions of Nozzles after cutting Nozzle Sleeves (See and Record at the Measurement Check Sheet).					 Check nozzle protrusions 
5	CRACK & LEAK TEST CYL.HEAD	Pressure test	- Air Pressure Test 1. Assemble plate and flange 2. Connect a hose to flange 3. Place the Cyl. Head in a water bath with hot water 4. Apply air pressure (3.0 ~ 3.5 kg/cm²) for approx 5 ~ 10 Min. 5. Check for any air leakage in the water. 6. Replace Cyl. Head if there are crack.	3.0 ~ 3.5 kg/cm ² for approx 5 ~ 10 Min. (82°C ~ 93°C)				  Check cylinder head with hot water & air pressure

ASSEMBLY CHECK SHEET CYLINDER HEAD ENGINE ALL SERIES	WO NO.		PART NAME	CYL. HEAD ASS'Y	DOCUMENT NO.	RC/QR04/EG/03/03
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	ENGINE S/N		START DATE /TIME		REVISION NO.	00
	MACHINE MODEL	GD 825-2	FINISH DATE / TIME		PAGE	2 of 2

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS & COMMENT)																						
					MEC	QA	SPV																							
6	VALVE & SPRING ASSEMBLY		- Make sure Cylinder Head free from Dirt and foreign material - Make sure all component that will assembly already clean 1. Align with casting mark on Cylinder Head and assembly Intake valve and Exhaust valve coat the valve steam and the face of the Valve thinly with engine oil (SAE 15W ~ 40) • Make sure of Valve (Intake and Exhaust) correct arrangement • Make sure the Valves are following to PSN Valve code					     																						
		PSN	<table><thead><tr><th>ENGINE MODEL</th><th>VALVE INTAKE</th><th>VALVE EXHAUST</th></tr></thead><tbody><tr><td>140 SERIES</td><td>I</td><td>CX / PX</td></tr><tr><td>170 SERIES</td><td>AN / BN</td><td>EX / X</td></tr><tr><td>125 SERIES</td><td>IN / BN</td><td>EX / BX</td></tr><tr><td>155 SERIES</td><td>I</td><td>E</td></tr><tr><td>170-3</td><td>IN</td><td>E</td></tr><tr><td colspan="3">ACTUAL</td></tr><tr><td></td><td></td><td></td></tr></tbody></table>	ENGINE MODEL	VALVE INTAKE	VALVE EXHAUST	140 SERIES	I	CX / PX	170 SERIES	AN / BN	EX / X	125 SERIES	IN / BN	EX / BX	155 SERIES	I	E	170-3	IN	E	ACTUAL								
ENGINE MODEL	VALVE INTAKE	VALVE EXHAUST																												
140 SERIES	I	CX / PX																												
170 SERIES	AN / BN	EX / X																												
125 SERIES	IN / BN	EX / BX																												
155 SERIES	I	E																												
170-3	IN	E																												
ACTUAL																														

INSPECTION RESULT :

ASSEMBLY BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)



CYLINDER HEAD CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

KOMATSU ENGINE

6D140-2

MACHINE MODEL

WO NO. :

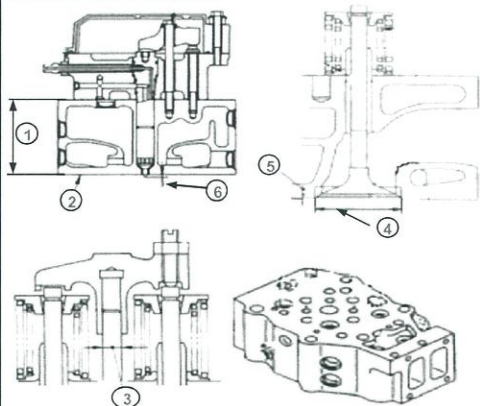
SUB ASS'Y SECTION

CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

JOB NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC/QA/EG/190
ENGINE MODEL	6D140-2	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	1
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 3

MEASURED BY		QUANTITY	
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STANDARD FIGURES



No	Measuring items			Std Tolerance (mm)	Repair Limit (mm)
1	Head Height			105.05 ~ 104.95	104. 65
2	Lower Surface Distortion			0,060	0,090
3	O.D. of Cross Head Guide			11.011 ~ 11.000	10,95
4	Valve Seat Insert Bore	In	Std	55.000 ~ 55.019	-
			0.25 OS	55.250 ~ 55.269	-
			0.50 OS	55.500 ~ 55.519	-
		Ex	Std	52.000 ~ 52.019	-
			0.25 OS	52.250 ~ 52.269	-
			0.50 OS	52.500 ~ 52.519	-
5	Valve Sink			0 ± 0.10	0.70
6	Nozzle Protrusion			4.8 ~ 5.6	Replace Nozzle or Cyl. Head

A.Result of Measurement

[illegible]

CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

JOB NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC/QA/EG/190
ENGINE MODEL	6D140-2	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	1
MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 3

7	ID Valve Guide (Use Tool G/NG)	In	1	NEW PARTS
			2	
		Ex	1	
			2	

[illegible]

Unit : mm

ENGINE MODEL	VALVE SPRING	STANDARD SIZE		REPAIR LIMIT
		FREE LENGTH	INSTAL LENGTH	LOAD (KG)
12V140 Series	INNER	72,00	46,00	19,70
	OUTER	81,70	51,00	38,70
6D140 Series	INNER	72,00	46,00	19,60
	OUTER	81,70	51,00	38,50
6D170-3	INNER	-	-	-
	OUTER	99,60	87,40	62,00
6D170 Series	INNER	59,30	47,00	17,40
	OUTER	64,80	37,50	84,80
6D170 HD465-5	INNER	73,80	48,00	23,30
	OUTER	77,00	52,00	48,70
6D125-2	INNER	-	-	-
	OUTER	75,20	56,00	47,70
6D155-4	INNER	-	-	-
	OUTER	88,00	51,90	65,50

[illegible]

CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

JOB NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC/QA/EG/190
ENGINE MODEL	6D140-2	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	1
MACHINE MODEL		FINISH DATE / TIME		PAGE	3 of 3

Method of measurement of depth, and measurement tool :
Visual inspection or use Dial gauge.

Judgement of Distortion(Flatness)

Measure clearance between Head and straight edge
Measuring point : Refer the right picture



Upper surface	Flatness	Remedy
	within 0.08mm	Use as it is
	More than 0.08mm	Reuse after surface grinding
lower surface	within 0.08mm	Reuse after surface grinding

Measurement of Distortion , and measurement tool
Straight edge, Clearance gauge

Measure flatness of the bottom surface of head

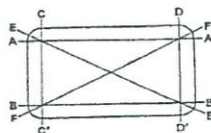


Fig. A



Fig. B



INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV. 00



PLAN REBUILD CENTER

CYLINDER HEAD

SALVAGING & MACHINING

CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

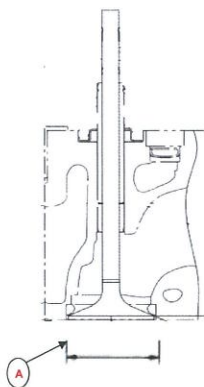


PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CYLINDER HEAD MACHINING AND MEASUREMENT CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0023
ENGINE MODEL	ALL SERIES	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE/TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



Standard size

ENGINE	VALVE SINK WITHOUT SPRING (A)		VACUM TEST
MODEL	STANDARD(mm)	REPAIR LIMIT(mm)	(BAR)
12V140	0 ± 0,10	0,70	- (6 ~ 8)
6D170-5	0 ± 0,10	0,80	
6D170E-3	0 ± 0,10	0,40	
6D140E-1	0 ± 0,10	0,70	
6D140E-5	0 ± 0,10	0,70	
6D125-1	0 ± 0,10	0,70	

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Salvaging			
Measuring			
TOTAL			

VISUAL CHECK	MECHANIC		QA	
	OK	NOT OK	OK	NOT OK
CRACK HAS BEEN CHECK				

No.	Measuring Item			R1		R2		R3		R4		R5		R6		RANDOM INSPECTION
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	
1	Valve Sink	In	1													
			2													
		Ex	1													
			2													
2	Vacuum Test	STD	IN													
		(6~8 Bar)	EX													
	Measuring Item			L1		L2		L3		L4		L5		L6		RANDOM INSPECTION
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	
3	Valve Sink	In	1													
			2													
		Ex	1													
			2													
4	Vacuum Test	STD	IN													
		(6~8 Bar)	EX													

INSPECTION RESULT :

DELIVERY BY MACHINING
DELIVERY DATE :
MACHINIST NAME :
SIGN

RECEIVING INSPECTION BY SUB ASSY
RECEIVED DATE :
MECHANIC NAME :
SIGN

Note : New Cyl Head & after finishing can be directly approval by QA (Quality Assurance)

REV. 00



PLAN REBUILD CENTER

IDLER GEAR AND SHAFT

MEASUREMENT CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

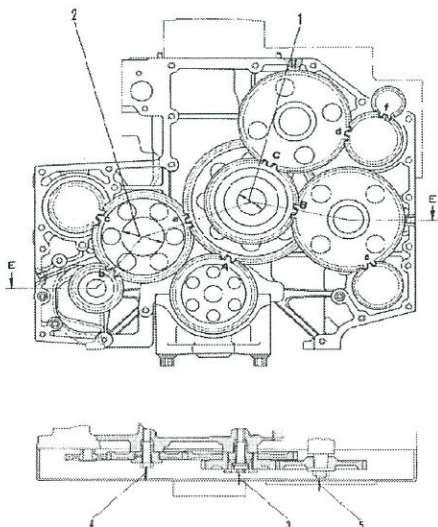


PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

SALVAGE IDLER GEAR AND SHAFT PROCESS INSPECTION CHECKSHEET

WO NUMBER		PART NAME	IDLER GEAR AND SHAFT	DOCUMENT NO.	
ENGINE MODEL	6D140 Series	PART NUMBER		PUBL. DATE	08/10/2009
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



Physical Data

No.	Part Number	Part Name	Qty	New	Reuse
1					
2					
3					
4					

MACHINIST NAME

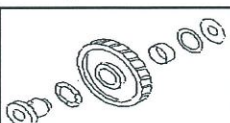
Standard size (unit mm)

DESCRIPTION	STANDARD DIMENSION		CLEARANCE	
	SHAFT	BUSHING	STANDARD	REPAIR LIMIT
ENGINE S/N : 10001 ~ 10368				
Main Idler Gear	47.614 ~ 47.601	47.685 ~ 47.630	0.016 ~ 0.084	0.20
Sub Idler Gear	47.614 ~ 47.601	47.685 ~ 47.630	0.016 ~ 0.084	0.20
ENGINE S/N : 10369 and up				
Main Idler Gear	55.915 ~ 55.970	55.986 ~ 55.999	0.016 ~ 0.084	0.20
Sub Idler Gear	49.945 ~ 50.000	49.984 ~ 49.971	0.016 ~ 0.084	0.20

VISUAL CHECK	MECHANIC		QA	
	OK	Not OK	OK	Not OK
CRACK HAS BEEN CHECKED				

MANHOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Measuring			
TOTAL			

INSPECTION SALVAGE PROCESS

SKETCH DRAWING	DIMENSION		ACTUAL DIAMETER						ACTUAL CLEARANCE			ROTATING CHECK (OK or NOT)		
			SHAFT			BUSHING								
			MEC	QA	SPV	MEC	QA	SPV	MEC	QA	SPV	MEC	QA	SPV
Main Idler Gear 	A	X							MIN					
		Y												
	B	X												
		Y								MAX				
	C	X												
		Y												
Sub Idler gear 	A	X							MIN					
		Y												
	B	X												
		Y								MAX				
	C	X												
		Y												

MECHANIC INSPECTION COMMENT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)

REV. 00



PLAN REBUILD CENTER

ROCKER ARM

MEASUREMENT & POLISHING

CHECK SHEET

KOMATSU ENGINE 6D140-2

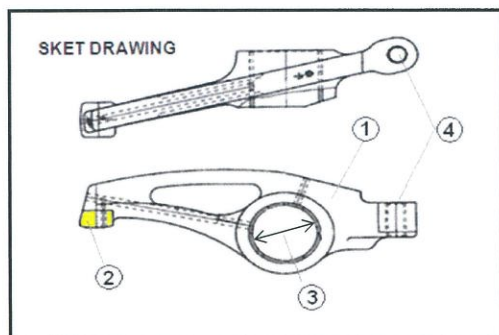
MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

ROCKER ARM POLISHING AND INSPECTION CHECK SHEET

JOB NUMBER		PART NAME	ROCKER ARM	DOCUMENT NO.	RC/QR03/EG/09/08
ENGINE MODEL	6D140E-2	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1



MACHINIST NAME	
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MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Polishing			
Inspection			
TOTAL			

[illegible][illegible]

REV. 00



PLAN REBUILD CENTER

ROCKER ARM SHAFT MEASUREMENT & POLISHING CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

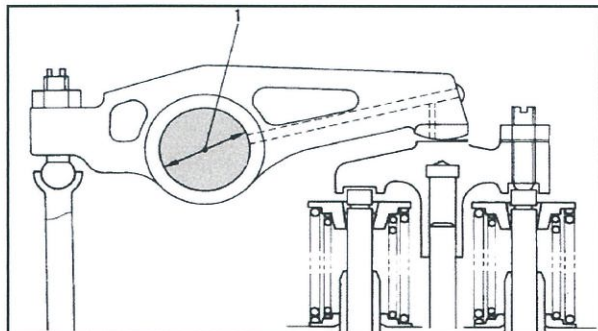
WO NO:

SUB ASS'Y SECTION

ROCKER ARM SHAFT MEASURING AND POLISHING CHECK SHEET

WO NUMBER		PART NAME	ROCKER ARM SHAFT	DOCUMENT NO.	RC/QR03/EG/09/08
ENGINE MODEL	6D140E-2	PART NUMBER		PUBL. DATE	05/02/2008
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



MACHINIST NAME

MAN HOURS UTILIZATION

PROCESS	START	FINISH	MAN HOURS
Machining			
Measuring			
TOTAL			

(Unit : mm)

ENGINE MODEL	STANDARD	STANDARD TOLERANCE	CLEARANCE LIMIT
6D140E-2	∅ 32.0000	∅ 32.0065 ~ 31.9935	0,13

INSPECTION MEASURING PROCESS

CYL.NO.	CHECK POSITION	1			2			3			Actual Clearance
		MAC	QA	SPV	MAC	QA	SPV	MAC	QA	SPV	
1	X - X'										
	Y - Y'										
2	X - X'										
	Y - Y'										
3	X - X'										
	Y - Y'										
4	X - X'										
	Y - Y'										
5	X - X'										
	Y - Y'										
6	X - X'										
	Y - Y'										

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MACHINIST (MACH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV. 00



PLAN REBUILD CENTER

CAM FOLLOWER & PUSHROD MEASUREMENT CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

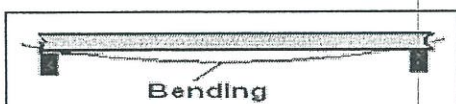
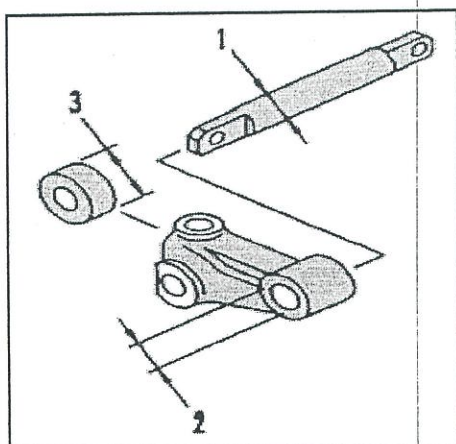
SUB ASS'Y SECTION

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	DOCUMENT NO.	
ENGINE MODEL	6D140 -2	PART NUMBER		PUBL. DATE	10/08/2009
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 3

MEASURED BY	
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STANDARD FIGURE



Unit : mm

ENGINE MODEL	CAM FOLLOWER RADIAL PLAY	
	STANDARD	Repair Limit
6D140 Series	0.077 ~ 0.150	0.151

Unit : mm

PUSH ROD BENDING STANDARD	
TIR max.	0.30

6D140 Series

Unit : mm

NO	CHECK ITEM	DIMENSION		
		SIZE	STD TOLERANCE	Repair Limit
1	OD of C.F. Shaft	22	21.979 ~ 21.966	21.96
2	ID of C.F. Shaft Hole	22	22.000 ~ 22.021	22.04
3	OD of Cam Roller	31.7	31.750 ~ 31.720	31.5

RADIAL PLAY CAM FOLLOWER						
CAM FOLLOWER	IN			EX		
	MECHANIC	GL	SPV	MECHANIC	GL	SPV
1						
2						
3						
4						
5						
6						

1	OUTSIDE DIAMETER OF CAM FOLLOWER SHAFT					
Out side Diameter of Cam Follower shaft	X			Y		
	MECHANIC	GL	SPV	MECHANIC	GL	SPV
1						
2						
3						
4						
5						
6						

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	DOCUMENT NO.	
ENGINE MODEL	6D140 -2	PART NUMBER		PUBL. DATE	10/08/2009
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	2 of 3

2	INSIDE DIAMETER OF CAM FOLLOWER SHAFT HOLE					
Inside Diameter of Cam Follower Shaft Hole	X			Y		
	MECHANIC	GL	SPV	MECHANIC	GL	SPV
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

3	OUTSIDE DIAMETER OF CAM ROLLER					
Out Side Diameter of Cam Roller	X			Y		
	MECHANIC	GL	SPV	MECHANIC	GL	SPV
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	DOCUMENT NO.	
ENGINE MODEL	6D140 -2	PART NUMBER		PUBL. DATE	10/08/2009
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	3 of 3

BENDING OF PUSH ROD						
PUSH ROD	IN			EX		
	MECHANIC	GL	SPV	MECHANIC	GL	SPV
1						
2						
3						
4						
5						
6						

MECHANIC INSPECTION COMMENT :

INSPECTED BY :	INSPECTED & APPROVED BY :	
MECHANIC (MECH)	QUALITY ASSURANCE (QA)	PROD. SUPERVISOR (SPV)



PT SAPTAINDRA SEJATI

PLANT REBUILD CENTER

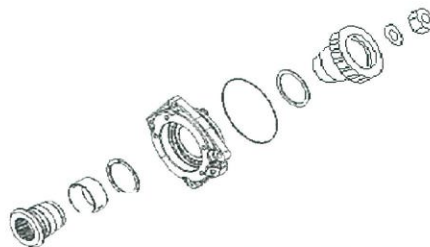
**COMPRESSOR DRIVE
ASSEMBLY CHECK SHEET
ENGINE KOMATSU**

6D140-2

MACHINE MODEL

WO NO:

SUB ASSY SECTION

ASSEMBLY CHECK SHEET		WO NUMBER		DOCUMENT NO		RC/QR04/EG/12/03	
COMPRESSOR DRIVE		COMP.MODEL		PUBLIC DATE			
SA6D140-2		COMP.S/N		REVISION NO			
		UNIT MODEL		PAGE			
PROCESS		START FINISH		ASSEMBLY BY			
NO	PART TO INSTALL OR FIT	CONDITION TOBE ASSMBLY	SPECIFICATION	SIGNATURE MECH GL/QA		SKETCH DRAWING OR PHOTHOES	
1	ALL PART	⚠ IMPORTANT ⚠ Confirm that Parts which have order do not DAMAGE,SCRATCH FREE FROM DIRT AND COROTION.					
2	BUSHING TO HOUSING	- Check Bushing seat and bushing from scratch,damage and corotion. - Install Bushing to housing . ⚠ IMPORTANT ⚠ Make sure surface bushing and bushing seat free from oil and dirt.	Using special tool for press (F2)				
3	TRUST PLATE AND SHAFT ASSY	- Confirm that shaft and trust plate free from dirt,crack,corrosion and damage. - Coat surface trust plate and shaft with oil. - Install trust plate() to housing. - Install shaft to housing.	SAE 15W - 40				
4	GEAR	- Confirm that gear free from scratch and pitting,corrosion and damage. - Install Gear() to the shaft () ⚠ IMPORTANT ⚠ Make sure inner gear and surface shaft free from dirt and oil.	Using special tool for press (F2)				
NOTE :							
ASSEMBLY BY		INSPECTED BY		APPROVED BY			
MECHANIC		Quality Assurance(QA)		GL/SPV PRODUCTION			

NOTE	
MECHANIC	GROUP LEADER (GL)



PT SAPTAINDRA SEJATI

PLANT REBUILD CENTER

PTO DRIVE

ASSEMBLY CHECK SHEET

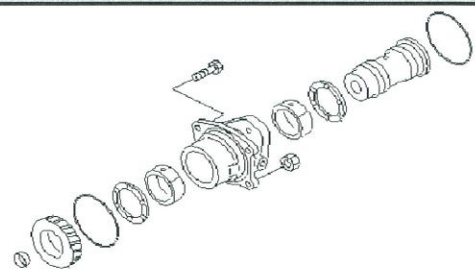
ENGINE KOMATSU

6D140-2

MACHINE MODEL

WO NO:

SUB ASSY SECTION

ASSEMBLY CHECK SHEET PTO DRIVE GEAR SA6D140-2		WO NUMBER		DOCUMENT NO		RC/QR04/EG/12/03	
		COMP.MODEL		PUBLIC DATE			
		COMP.S/N		REVISION NO			
		UNIT MODEL		PAGE			
PROCESS		START FINISH		ASSEMBLY BY			
NO	PART TO INSTALL OR FIT	CONDITION TOBE ASSMBLY	SPECIFICATION	SIGNATURE		SKETCH DRAWING OR PHOTHOES	
				MECH	GL/QA		
1	ALL PART	⚠ IMPORTANT ⚠ Confirm that Parts which have order do not DAMAGE,SCRATCH FREE FROM DIRT AND COROTION.					
2	BUSHING TO HOUSING	• Check Bushing seat and bushing from scratch,damage and corotion. • Install Bushing to housing . ⚠ IMPORTANT ⚠ Make sure surface bushing and bushing seat free from oil and dirt.	Using special tool for press (F2)				
3	TRUST PLATE AND SHAFT ASSY	• Confirm that shaft and trust plate free from dirt,crack,corrosion and damage. • Coat surface trust plate and shaft with oil.	SAE 15W - 40				
4	GEAR	• Confirm that gear free from scratch and pitting,corrosion and damage. • Install Gear() to the shaft () ⚠ IMPORTANT ⚠ Make sure inner gear and surface shaft free from dirt and oil. • Install plug to shaft,tap with blow hammer carefully.	Using special tool for press (F2) Using special tool				
NOTE :							
ASSEMBLY BY		INSPECTED BY		APPROVED BY			
MECHANIC		Quallity Assurance(QA)		GL/SPV PRODUCTION			

MEASUREMENT CHECK SHEET
PTO DRIVE GEAR
SA6D140-2

WO NUMBER
COMP.MODEL
COMP.S/N
UNIT MODEL

DOCUMENT NO
PUBLIC DATE
REVISION NO
PAGE

RC/QR03/EG/11/04

PROCESS

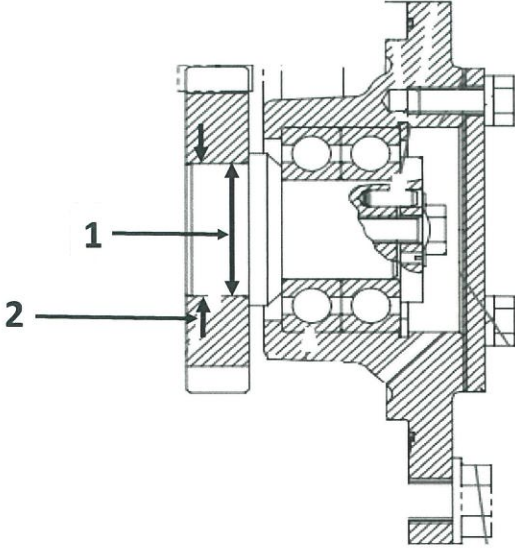
START
FINISH

MEASUREMENT BY

NO

PART TO MEASURING OR
INSPECT

STANDARD VALUE TABLE



1

PTO DRIVE GEAR

unit:mm

NO	CHECK ITEM	CRITERIA		ACTUAL
		STANDART SIZE	POSITION	
1	Outside diameter of shaft	40,060 ~ 40,076	1	X
			1	Y
			2	X
			2	Y
2	Inside Diameter of Gear	40,000 ~ 40,025	1	X
			1	Y
			2	X
			2	Y
3	Interference between OD shaft & ID gear	0,030 ~ 0,076	1	MIN
			1	MAX
			2	MIN
			2	MAX
4	VISUAL CHECK	OK/NOT OK		

NOTE

MECHANIC

GROUP LEADER (GL)



PT SAPTAINDRA SEJATI

PLANT REBUILD CENTER

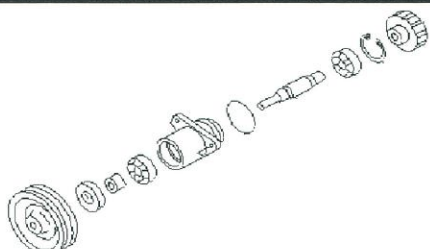
**FRONT PTO PULLEY
ASSEMBLY CHECK SHEET
ENGINE KOMATSU**

6D140-2

MACHINE MODEL

WO NO:

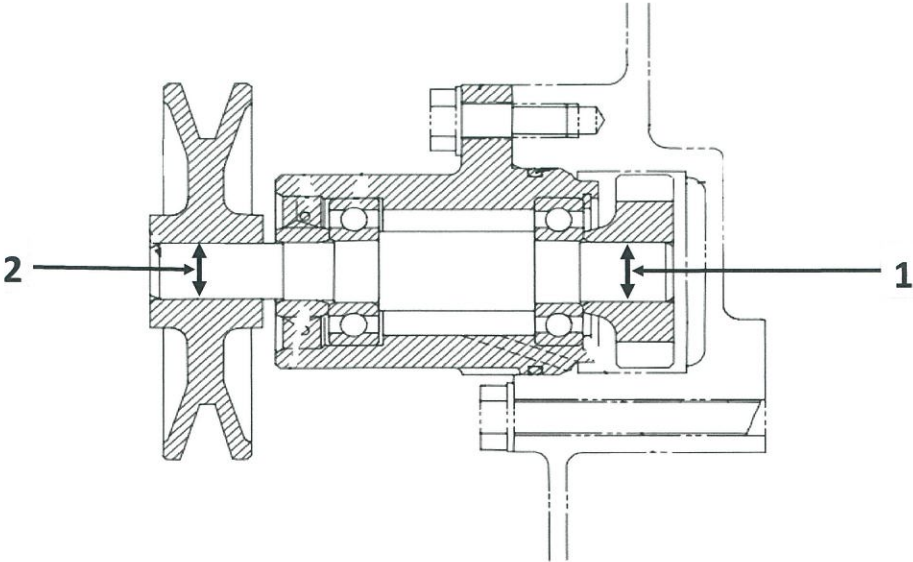
SUB ASSY SECTION

ASSEMBLY CHECK SHEET		WO NUMBER	DOCUMENT NO		RC/QR04/EG/12/03	
FRONT PTO PULLEY		COMP.MODEL	PUBLIC DATE			
SA6D140-2		COMP.S/N	REVISION NO			
		UNIT MODEL	PAGE			
PROCESS	START FINISH		ASSEMBLY BY			
NO	PART TO INSTALL OR FIT	CONDITION TO BE ASSEMBLY	SPECIFICATION	SIGNATURE		SKETCH DRAWING OR PHOTOES
			MECH	GL/QA		
1	ALL PART	⚠ IMPORTANT ⚠ Confirm that Parts which have order do not DAMAGE, SCRATCH FREE FROM DIRT AND CORROSION.				
2	SHAFT TO BEARING	- Check surface bearing and shaft from scratch, damage and corrosion. - Install Shaft to bearing. ⚠ IMPORTANT ⚠ hold inner race from bearing at pressing shaft.	Using special tool for press (F2)			
3	BEARING, SHAFT TO HOUSING	- Confirm that housing free from dirt, crack, corrosion and damage. - Install bearing and shaft to housing, and install the other bearing from back side housing. ⚠ IMPORTANT ⚠ Bearing seat in housing must be clean and no scratch	Using special tool for press (F2)			
4	SNAPRING, SLEAVE AND OIL SEAL	- Install snapring () to the housing. ⚠ IMPORTANT ⚠ Make sure groove snapring seen and can be installed. - Install Sleeve () and Install oil seal () - Apply Gasket sealant to the outer face oil seal, then tap with plastic hammer to install oil seal.	LG - 6 Using plastic hammer			
5	GEAR ASSY	- Make sure Gear assy free from scratch, pitting and damage. - Install Gear () to the shaft.	Press Fit			
6	PULLEY ASSY	- Make sure pulley assy free from worn out and damage. - Install Shaft to Gear.	Press Fit			
NOTE :						
ASSEMBLY BY INSPECTED BY APPROVED BY						
MECHANIC Quality Assurance(QA) GL/SPV PRODUCTION						

MEASUREMENT CHECK SHEET FRONT PTO PULLEY SA6D140-2	WO NUMBER		DOCUMENT NO	RC/QR03/EG/11/04
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

PROCESS	START FINISH		MEASUREMENT BY		
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NO	PART TO MEASURING OR INSPECT	STANDARD VALUE TABLE
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unit:mm

1 FRONT PTO PULLEY

NO	CHECK ITEM	CRITERIA		ACTUAL
		STANDART SIZE	POSITION	
1	Outside diameter of shaft	19.002 ~ 19.015	1	X
			1	Y
			2	X
			2	Y
	Inside Diameter of Gear	18.972 ~ 18.951	1	X
			1	Y
			2	X
			2	Y
	Interference between OD shaft & ID gear	0.030 ~ 0.064		
2	Outside diameter of shaft	18.002 ~ 18.015	1	X
			1	Y
			2	X
			2	Y
	Inside Diameter of Pulley	17.972 ~ 17.951	1	X
			1	Y
			2	X
			2	Y
	Interference between OD shaft & ID pulley	0.030 ~ 0.064		
3	VISUAL CHECK	NO SCRATCH		

NOTE

MECHANIC

GROUP LEADER (GL)

REV. 01



PLAN REBUILD CENTER

WATER PUMP

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

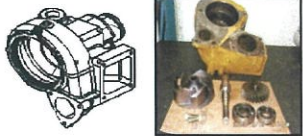
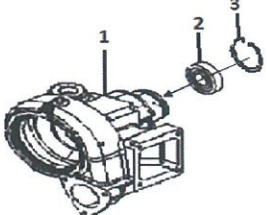
ASSEMBLY CHECK SHEET

WATER PUMP

6D140 SERIES

WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	
ENGINE MODEL		PART NUMBER		PUBLIC DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 3

ASSEMBLY BY		All dimension in millimetres
		All torques in Kilogram metres

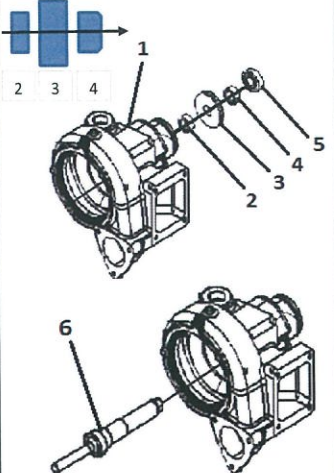
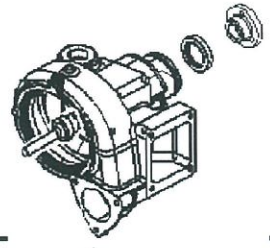
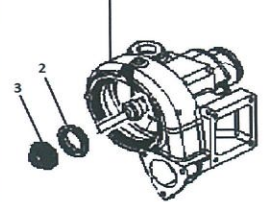
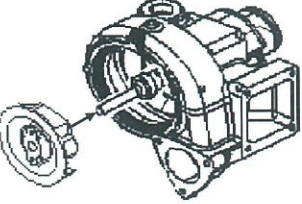
NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
1	Pump Body		Pump body must be free from crack, dent and rush Clean All Parts throughly and make sure that all burns have been remove. Always check the contact of the following parts before assembly. See checking contact between parts. Marking with paint maker, for all of bolts and nuts after tightening by torque wrench	See maintenance standard for standard dimension			
2	Housing		Housing must be free from crack, dents and rust All threads are in good condition and clean Make sure that face (clearance area) is in good and clean				
3	Other Parts		Recheck condition of bearings, oil seal, shaft, impeller, water seal, gear, snap ring, bolts and washers should be free from crack, rust, dents, scratch and pitting	Refer and follow PSN			
4	Install Oil Seal to housing		Oil seal seat must be clean and dry (free from oil, fuel or water) Coat the out side of oil seal with LG 1215. Insert oil seal using special tools Coat the seal lip with grease sufficiently	Use oil seal installer G2-LI			
5	Insert Bearing to housing		Bearings(2) and bearing seat(1) must be clean. Coat Few oil in out race bearing. Insert Big bearing to housing using special tools. Make sure that bearings center to housing(bearing seat)when assembly. Install snapping (3).to groove until fixed.	Use special Tools and Press by Hyd.Jack			

ASSEMBLY CHECK SHEET

WATER PUMP

6D140 SERIES

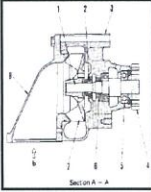
WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	
ENGINE MODEL		PART NUMBER		PUBLIC DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 3

NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
6	Alignment drive gear assy and insert small bearing assy		Align spacer (2), drive gear (3) & Spacer with groove (4). Coat Few oil in out race bearing. Insert Small bearing to housing using special tools. Insert shaft ass'y (6) to the drive gear from back side housing. Hold small bearing with special tool when install shaft assy. Make sure that shaft assy is centering to body when shaft insert to spacer and drive gear. CAUTION!! Recheck and make sure that oil seal, spacer and gear An is hit or n't damage when inserting the shaft Check shaft from smooth rotation. Standard of interference between Gear and Shaft Act. Interference : mm	Use special Tools (Shaft Installer) 0.025 ~ 0.07 mm			
10	Insertion Seal and hub		Insert Seal and Hub.				
8	Insertion Water seal to housing		Make sure that the condition of water seal is free from scratch, crack, pitting, dents, corrosive, worn, and etc. Here is critical point, be careful Coat the outer side of water seal with adhesive (contact with housing) Insert water seal using special tools Coat the rubber surface water seal with "soap" (contact with impeller)	Follow and refer to the PSN LOCTITE 609 (green loctite) Toilet Soap (sabun)			
9			Insert the impeller and Make sure that is fit in the groove. Coat the hole of the impeller with small quantity of BRAKE FLUID be carefull when insert the Impeller Insert impeller to shaft with Special tool. Standard of interference between impeller and Shaft Act. Interference : mm Clearance between impeller and connector / housing Standard Clearance Act. Clearance = mm Insert the impeller to the shaft until contact to the end of the shaft.	Use special tool and hdy jack 0.025 ~ 0.068 mm 0.23 ~ 0.66 mm			 



ASSEMBLY CHECK SHEET WATER PUMP 6D140 SERIES

WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	
ENGINE MODEL		PART NUMBER		PUBLIC DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	3 of 3

NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
11	General Inspection		✓ : Check - Check Shaft for smooth rotation OK - Recheck position of snap ring OK - Check loose bolts from mistake OK - Check all parts from improper installation OK - Check and keep completion parts OK				 1. Pump shaft 2. Water seal 3. Pump body 4. Water pump drive gear (No. of teeth: 25) 5. Ball bearing 6. Oil seal 7. Impeller 8. Pump cover  a. From thermostat b. From radiator (water inlet) c. To oil cooler (water outlet)
12	End play water pump		Measure End play of water pump shaft before delivery to Main line 	Standard End play 0			

MECHANIC ASSEMBLY COMMENT

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV. 01



PLAN REBUILD CENTER

AIR COMPRESSOR

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

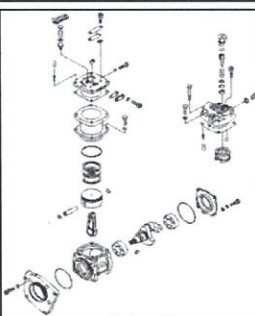
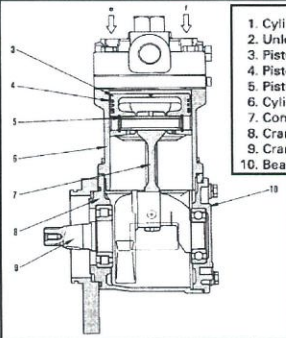
SUB ASS'Y SECTION

AIR COMPRESSOR ASSEMBLY CHECK SHEET KOMATSU ENGINE 6D140-2

JOB NUMBER		PART NAME	AIR COMPRESSOR	DOCUMENT NO.	RC/QR04/EG/09/02
ENGINE MODEL	6D140-2	PART NUMBER		PUBL. DATE	01/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL	GD 825A-2	FINISH DATE / TIME		PAGE	

ASSEMBLY BY	
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1 of 3

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
1	All Parts		Make sure all parts are ready to used (No Scratch and Worn)					
2	Flange and Crankcase		Fit O-ring to crankcase then assembly Flange tighten with mounting bolts ★ Coat thread with thread locker (Three Bond 1360 or LT 2) Fit outer race front bearing to crankcase crankcase ★ Coat outer race bearing with bearing locktite or equivalent. Make sure that surface flange must be free from worn and scratch. Install flange front that has a lubrication hole and more thick than rear side of flange (For 12V140-1 Series)					 1. Cylinder head 2. Unloader valve 3. Piston 4. Piston ring 5. Piston pin 6. Cylinder 7. Connecting rod 8. Crank case 9. Crankshaft 10. Bearing cover a. Air inlet b. Air outlet c. Unload pressure (from air governor) d. Oil inlet e. Water inlet or outlet f. Water inlet or outlet
3	Crankshaft and Con.Rod		Fit front bearing to crankshaft smoothly ★ Use hydraulic pressure Insert crankshaft in to crankcase, insert big end connecting rod from cylinder mounting fortian then push in crankshaft (clearance between bearing outer race is zero or less) Make sure the position of connecting rod is correct Press fit ball bearing inner to Crankshaft ★ Use hydraulic pressure	$\pm 100 \text{ Psi}$ $\pm 100 \text{ Psi}$				Be careful when fit of bearing crankshaft. Con rod code (align to front side)
4	Cover Bearing		Fit O-ring then press fit bearing cover to ball bearing ★ Hit the bearing cover with plastic hammer slowly ★ Tightening the mounting bolts of the Cover Actual : kgm Spray engine oil bearing and then rotate the crankshaft slowly	2.5 ~3.0 kgm				Make sure that the O-ring is not drop Make sure that the crankshaft rotation is in normal condition

AIR COMPRESSOR ASSEMBLY CHECK SHEET KOMATSU ENGINE 6D140-2

JOB NUMBER		PART NAME	AIR COMPRESSOR	DOCUMENT NO.	RC/QR04/EG/09/02
ENGINE MODEL	6D140-2	PART NUMBER		PUBL. DATE	01/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL	GD 825A-2	FINISH DATE / TIME		PAGE	

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
5	Piston ring Piston Pump and Cyl. Assembly		Install oil ring and piston rings to piston ★ Install Piston Ring with special tool (Ring Expander) Assembly piston rings with the stamp ring surface facing up, and make sure that the end gaps of the rings are in different places and same condition with assembly engine piston ring Oil ring is not top or bottom surface coat cylinder inner with engine oil (SAE 15W ~ 40) ★ Install Oil Ring with special tool (Ring Expander) Insert the piston assembly to cylinder liner from top side until to the side and align piston pin hole and small end of connecting rod then push in piston pin and secure bolts end with snap rings ★ Piston ring coat with engine oil (SAE15W~40) Fit O-ring to cylinder liner and assembly cylinder to crankcase and tighten with mounting bolts Actual : kgm	End gaps of ring should be facing up 2.7~3.3 kgm				Make sure that the ring piston arrangement are in correct position 1. Top ring code : R and B 2. Second ring code : R Be careful when insert piston assembly to cylinder liner from crack / broken of ring piston Check piston code (), does not face plate name or code (L) must face plate name Make sure that cylinder liner must be correct specification unit model
6	Cylinder Head Group	Delivery Valve Assy	Install delivery valve, delivery valve guide and stagger then tighten with hexagon socket head screws ★ Coat thread with thread locker (Three Bond 1360 or LT 2) Actual : kgm	0.4~0.5 kgm				Be careful not to let the delivery valve and delivery valve guide overlap Be careful not to get the thread tighten on the delivery valve
		Inlet Valve Assy	Assembly inlet valve seat to cylinder head then tighten mounting bolt Thread and washer must be coat with thread tighten ★ Coat thread with thread locker (Three Bond 1360 or LT 2) Actual : kgm	0.54~0.66 Kgm				Be careful when install gasket head with liquid must face cylinder head plate
		Plate	Check plate must be free from worn and scratch Tighten plug or stopper valve plate thread must be coat with thread tighten Actual : kgm Insert washer, inlet valve spray and inlet plate to plate then assembly gasket and O ring and tighten push inlet valve plate from in loader valve end to check movement. Check that the tip of the hexagon socket head screw does not protrude from the surface of plate	0.54~0.66 Kgm				Tighten with Hexagon socket head screw 

AIR COMPRESSOR ASSEMBLY CHECK SHEET KOMATSU ENGINE 6D140-2

JOB NUMBER		PART NAME	AIR COMPRESSOR	DOCUMENT NO.	RC/QR04/EG/09/02
ENGINE MODEL	6D140-2	PART NUMBER		PUBL. DATE	01/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL	GD 825A-2	FINISH DATE / TIME		PAGE	

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
		Unloader Valve	Assembly O-ring to rod , insert unloader valve spring and washer then install snap ring Install un loader valve body to cylinder head. Thread of unloader valve must be coat with thread tighten Fit cylinder head gasket to top surface of cylinder head to cylinder then tighten with mounting bolts Tighten the bolts slowly and securely					
7	Coupling and Nut		Fit key to groove of crankshaft and install spline shaft, tighten the nut slowly Actual : kgm	25~28 kgm				Use special tools 
8	Final Inspection		Recheck all bolts from loaser and give recheck marks Turn crankshaft slowly and make sure the crankshaft condition is good. For normal condition give stand code (Job No.) Playing all holes and pack the compressor Check condition un loader valve group are worked	White marks				

[illegible]

ASSEMBLY BY

MECHANIC (MECH)

INSPECTED & APPROVED BY

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV. 01



PLAN REBUILD CENTER

OIL COOLER

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

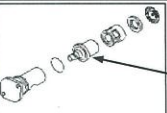
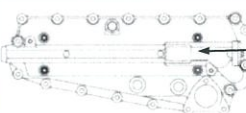
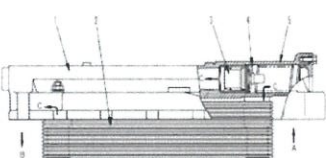
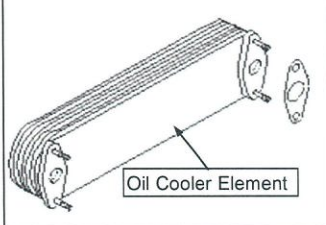
MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

ASSEMBLY CHECK SHEET OIL COOLER KOMATSU ENGINE 6D140-5	JOB NUMBER		DOCUMENT NO	RC/QR04/EG/06/03
	ENGINE MODEL	6D140-2	PUBLISHED DATE	
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL	GD 825A-2	PAGE	1 of 1

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	SPV	QA	
1	Core Element		- Test and check flow on core element. - Test and check leakage on core element	Oil cooler flusher				 
2	Thermostat		- Check cleanliness and damage crack on thermostat. - Test and check function of thermostat.					 Thermostat
3	Housing		- Check cleanliness and damage crack on element housing. - Check all threads on element housing					 Oil Cooler Housing
4	Assembly of Thermostat		- Install seal on housing thermostat. - Install thermostat on housing oil cooler - Install o ring on cover thermostat - Install cover thermostat on oil cooler housing . - Install bolt then tighten. Actual : Kgm	3.2 ±0.3Kgm				
5	Assembly of Oil Cooler Element		- Fit O ring or gasket on housing - Install washer and nut then tighten. - Tighten Torque : Actual : Kgm	Apply LG-6 M10 = 2.8~3.8Kgm M12 = 5.6~6.6Kgm (Komatsu assembly manual)				 Oil Cooler Element

Comment :

ASSEMBLY BY

INSPECTION & APPROVED BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV. 01



PLAN REBUILD CENTER

OIL COOLER

WATER LEAK TEST CHECK SHEET

KOMATSU ENGINE 6D140-2

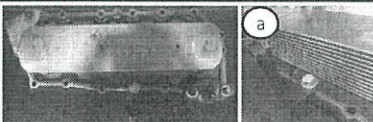
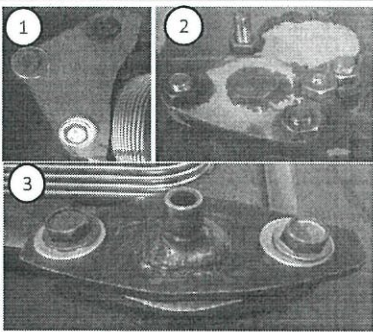
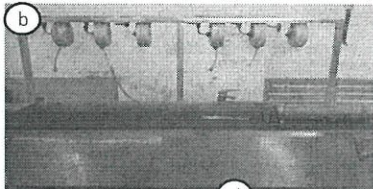
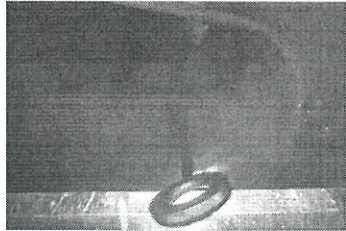
MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

WATER LEAK TEST CHECK SHEET OIL COOLER ASSY KOMATSU ENGINE 6D140 SERIES	WO NUMBER		Eks UNIT CODE	
	ENGINE MODEL	6D140 SERIES	PUBLISHED DATE	22-Jan-15
	ENGINE S/N		REVISION NO	01
	MACHINE MODEL		DOCUMENT NO	

PROCESS	START		TEST BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Core Element		- Check Element normal condition. (broken from bend,rust etc) - Install hook/hanger (a).	Visual check				
2	Plug,adapter and Air inlet connector		- Install plug on housing (1). - Install adapter and use O ring (2) - Install inlet air connector and use sealant or rubber for this connector (3)					
3	Water leak tester and indicator		- Prepare water leak tester tool (b) - Setting water temperature at standar range '(c). Actual : °C - Setting air pressure on Regulator valve (d) Actual : Kg/cm	Water leak tester tool STD range tmpratur 75° ~ 90°C (Shop manual) 3 ~ 4 kg/cm (Reman manual)				  
4	Water leak test oil cooler		- Install coupler and test oil cooler assy - Time for test Actual : Minutes	5 Minutes (Reman manual)				

Comment :

ASSEMBLY BY	INSPECTION & APPROVED BY
MECHANIC (MEC)	QUALITY ASSURANCE (QA) PROD. SUPERVISOR (SPV)

REV. 01



PLAN REBUILD CENTER

OIL COOLER

AIR LEAK TEST CHECK SHEET

KOMATSU ENGINE 6D140-2

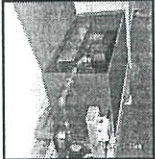
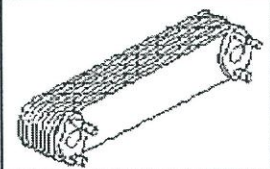
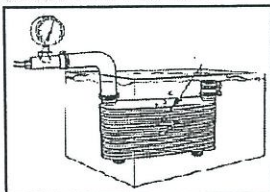
MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

AIRLEAK TEST CHECK SHEET OIL COOLER ELEMENT KOMATSU ENGINE 140 SERIES	WO NUMBER		DOCUMENT NO	QR00-SA-008-01
	ENGINE MODEL	6D140 SERIES	PUBLICATION DATE	
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START DATE / TIME	TESTED BY
	FINISH DATE / TIME	

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Oil Cooler Element	Air Leak test	- Inspect Element for cracks by visual check. No Crack	√ = CHECK				 
2	Test pressure		Check for Crack by Air pressure test for Element a). Air pressure: Actual : Kgm/cm2 b). Water Temperature Actual : °C c). Time for test Actual : Minutes No Leakage	5~7 kg/cm2 (Reman Manual) 70°C (Reman Manual) 5 minutes (Reman Manual) √ = CHECK No Leakage				 

COMMENT :

TESTED BY	INSPECTION & APPROVED BY	
_____ MECHANIC (MECH)	_____ QUALITY ASSURANCE (QA)	_____ PROD. SUPERVISOR (SPV)

REV. 01



PLAN REBUILD CENTER

TURBOCHARGER

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE 6D140-2	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL	SA6D140-2	PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

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ASSEMBLY BY	All dimensions in millimetres
	All torques in Kilogram metres

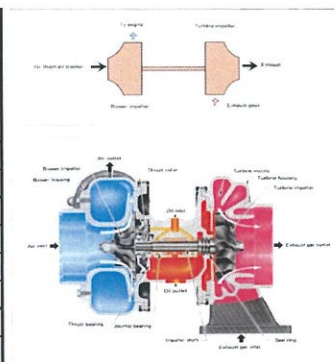
Notes :

- Before assembly parts, wash and clean the parts, allow them to dry by air blow, and apply engine oil to the sliding parts.
- Remove burrs completely from the parts (Pay attention to seal ring grooves, in particular).
- When reassembling the rotor assembly after the dynamic balance check has been made, be sure to align all the match marks stamped on the component parts (Turbine Shaft : Nut : Blower Impeller : Flinger and Collar).

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
1	Turbine Rotor	Seal ring Groove	Fit seal ring into Turbine rotor. Apply a coat of engine oil (SAE #30) to the side of seal ring and install the seal ring in such a way that the gap of each ring is positioned at 90° apart from that of the neighboring ring or 45° apart from oil fillet port on the center housing.					Position of 1st ring Position of 2nd ring

Thickness of Seal Ring

Thickness of Sawring					
	Standard size		Tolerance Width	Repair Limit Width	Actual
KTR 100	Turbine Side	2,2	-0,08	1,85	
			-0,1		
	Blower Side	1,95	-0,08	1,85	
			-0,1		
			4	-	-
KTR 110	Turbine Side	2,2	-0,08	1,85	
			-0,1		
	Blower Side	1,95	-0,08	1,45	
			-0,1		
KTR 130		3	-0,08	2,85	
			-0,1		
KTR 150		3	-0,08	2,85	
			-0,1		



No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
2	Shield		Install shield into the Turbine Rotor					
3	Center Housing	Seal ring seat	Install center housing on the Turbine When installing the center housing, take care to prevent the seal ring from slipping to installation					
4	Journal Bearing	Journal bearing surface	Install journal bearing Apply sufficient amount of engine oil to the inside, outside and both sides of the journal Bearing prior to installation					

Measurement of Journal Bearing,Center Housing & Wheel Shaft

	Standard Size	Tolerance		Repair limit		Actual	
		Shaft	Hole	Shaft	Hole	Shaft	Hole
KTR 100							
Outside diameter of journal bearing	25	-0,04	0,02	24,91	25,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	17	-0,03	0,02	16,94	17,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE 6D140-2	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL	SA6D140-2	PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

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KTR 110							
Outside diameter of journal bearing	25	-0,04	0,02	24,91	25,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	17	-0,03	0,02	16,94	17,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

KTR 130							
Outside diameter of journal bearing	30	-0,04	0,02	29,91	30,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	20	-0,03	0,02	19,94	20,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

KTR 150							
Outside diameter of journal bearing	30	-0,04	0,02	29,91	30,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	20	-0,03	0,02	19,94	20,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
5	Thrust		- Install thrust collar - Make sure that the thrust collar is free from foreign material on both sides, apply engine oil (SAE #30) both sides of the collar, and install that collar with the match mark aligning with marking mark stamp on the turbine shaft. - Drive a dowel pin in the center housing.					
6	Thrust Bearing		- Install thrust bearing with the dowel pin fitted in position - Before installing the trust Bearing, apply engine oil (SAE #30) to both sides of the oil hole					

Thickness of thrust bearing

KTR 100	Standard size	Tolerance Width	Repair limit Width	Actual Width
	5	-0,08 -0,11	4,86	
KTR 110	Standard size	Tolerance Width	Repair limit Width	Actual Width
	5	-0,08 -0,11	4,86	
KTR 130	Standard size	Tolerance Width	Repair limit Width	Actual Width
	6	-0,08 -0,11	5,86	
KTR 150	Standard size	Tolerance Width	Repair limit Width	Actual Width
	6	-0,08 -0,11	5,86	

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE 6D140-2	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL	SA6D140-2	PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

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No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
7	Flinger	One seal ring	Fit seal ring into flinger					
			Apply engine oil to the sides of the seal ring					
			Install flinger into insert (back plate)					
			Let the gap of the seal ring on the flinger					
			Direct to the oil filler port on the center housing					
8	Insert (Back Plate)	Two seal rings	Let the gap of the seal ring in such away that the gap of each ring is positioned at 90° apart from from that of the Neighboring ring or at 45° apart from oil filler port on the center housing.					
			For KTR 100 , KTR 130					
			• Fit an O-ring and install insert					
			• Tighten the insert mounting bolts and then bend the lock washer					
			Act. KTR 100 = kgm	KTR 100	1.25 ± 0.25			
			Act. KTR 130 = kgm	KTR 130	3.15 ± 0.35			
			For KTR 110, KTR 150					
			• Fit an O-ring on housing center					
			• Install back plate onto Turbine Rotor until contact the center housing and make sure that flinger with the match mark aligning with the marking mark stamp on the turbine shaft.					
			• Tighten the back plate mounting bolts and bend the lock washer.					
			Act. KTR 110 = kgm	KTR 110	3.15 ± 0.35			
9	Blower Impellers		Install blowers impellers onto the turbine rotor in the following manner immerse blower impeller in the oil (SAE #30) that is heated at 140° to 160° C for 5 to 20 minutes or immerse Blower					   
			Impellers with hot air gun for 5-10 minutes.					
			Install the blower impellers into the turbine rotor, aligning the match mark with the match mark stamp on the turbine shaft.					
			Apply engine oil to the inner face of the blower impeller and install the impeller without using excessive force					
			When tightening the nut, take care not allow any match marks to be out of alignment.					
			Make sure that nut with the match mark aligning with the marking mark stamp on the turbine shaft and blower impeller.					

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE 6D140-2	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL	SA6D140-2	PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

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No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
			Tightening torque :					
			Act. KTR 100 = kgm	KTR 100	4.25 ± 0.25			
			Act. KTR 110 = kgm	KTR 110	4.5 ± 0.25			
			Act. KTR 130 = kgm	KTR 130	8.5 ± 0.5			
			Act. KTR 150 = kgm	KTR 150	9.5 ± 0.5			
			Check End Play					
			Actual KTR 100 : = mm	KTR 100	Standard Repair limit	0.06 to 0.11 0,18		
			Actual KTR 110 : = mm	KTR 110	Standard Repair limit	0.08 to 0.13 0,18		
			Actual KTR 130 : = mm	KTR 130	Standard Repair limit	0.08 to 0.13 0,18		
			Actual KTR 150 : = mm	KTR 150	Standard Repair limit	0.08 to 0.13 0,18		
			Check Radial Play					
			Actual KTR 100 : = mm	KTR 100	Standard Repair limit	0.20 to 0.44 0,6		
10	Housing	Turbine Housing	Take care not to damage the blower impeller and turbine impeller when installing the blower housing and turbine housing					
			For KTR 110 , KTR 150 Install the Turbine housing into the center housing and the rotor assembly. Apply a thin coat of anti seiz compound to the flange section of turbine housing. Install lock plate, tighten mounting bolt. Apply a thin coat of anti seiz compound to the thread of bolt.					
			Tighten mounting bolt :					
			Act. KTR 110 = kgm	KTR 110	5.0 ± 0.5			
			Act. KTR 150 = kgm	KTR150	5.75 ± 0.25			
			Install blower housing, aligning it with the center housing and the rotor assembly. Apply a thin coat of Grease to the flange section.					
			For KTR 100 , KTR 130 Install the blower housing and turbine housing into the center housing and the rotor assembly.					
			Install V-Band clamp and tighten then torque of the V-Band locknut.					
			Act. KTR 100 = kgm	KTR 100	0.9 ± 0.1			
			Act. KTR 130 = kgm	KTR 130	0.9 ± 0.1			

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE 6D140-2	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL	SA6D140-2	PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

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No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS															
					MEC	QA	SPV																
			Install V-Band clamp and tighten then torque of the V-Band locknut.					 															
			<table border="1"> <tr> <td>Act. KTR 110 =</td> <td>kgm</td> <td>KTR 110</td> <td>0.9 ± 0.1</td> </tr> <tr> <td>Act. KTR 150 =</td> <td>kgm</td> <td>KTR150</td> <td>0.9 ± 0.1</td> </tr> </table>	Act. KTR 110 =	kgm	KTR 110	0.9 ± 0.1		Act. KTR 150 =	kgm	KTR150	0.9 ± 0.1											
Act. KTR 110 =	kgm	KTR 110	0.9 ± 0.1																				
Act. KTR 150 =	kgm	KTR150	0.9 ± 0.1																				
			Measure the clearance in the radial direction in the following manner.																				
			Depress the end of the rotor in the radial direction with a finger. Measure the clearance between the blower impeller and the housing by using filler gauge. Tolerance (Min.) : <table border="1"> <tr> <td>Act. KTR 100 =</td> <td>mm</td> <td>KTR 100</td> <td>0,15</td> </tr> <tr> <td>Act. KTR 110 =</td> <td>mm</td> <td>KTR 110</td> <td>0,2</td> </tr> <tr> <td>Act. KTR 130 =</td> <td>mm</td> <td>KTR 130</td> <td>0,2</td> </tr> <tr> <td>Act. KTR 150 =</td> <td>mm</td> <td>KTR 150</td> <td>0,2</td> </tr> </table>	Act. KTR 100 =	mm	KTR 100	0,15		Act. KTR 110 =	mm	KTR 110	0,2	Act. KTR 130 =	mm	KTR 130	0,2	Act. KTR 150 =	mm	KTR 150	0,2			
Act. KTR 100 =	mm	KTR 100	0,15																				
Act. KTR 110 =	mm	KTR 110	0,2																				
Act. KTR 130 =	mm	KTR 130	0,2																				
Act. KTR 150 =	mm	KTR 150	0,2																				
																							

AFTER COMPLETING THE ASSEMBLY, ROTATE THE ROTOR ASSEMBLY WHILE DEPRESSING IT LIGHTLY AND MAKE SURE THAT THE ROTOR CAN BE ROTATED SMOOTHLY WITHOUT BINDING.

[illegible]

ASSEMBLY BY	INSPECTION & APPROVAL BY :
MECHANIC (MEC)	QUALITY ASSURANCE (QA)
	GL/SPV PRODUCTION

REV. 00



PLAN REBUILD CENTER

TENSION PULLEY

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

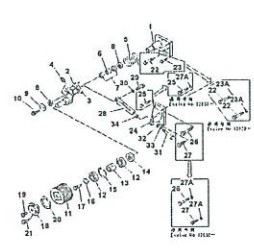
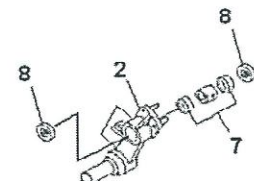
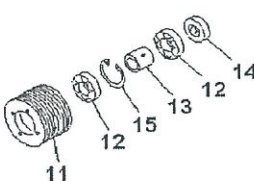
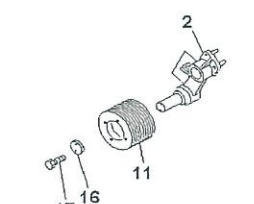
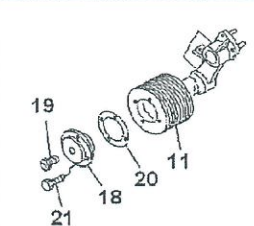
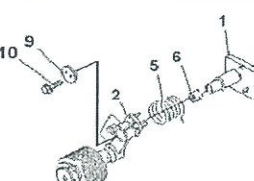
MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

ASSEMBLY CHECK SHEET TENSION PULLEY ENGINE 6D140-2 SERIES	WO NUMBER		DOCUMENT NO	
	ENGINE MODEL		PUBLISHED DATE	20/10/2009
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 2

PROCESS	START DATE / TIME		ASSEMBLY BY	
	FINISH DATE / TIME			

NO.	PARTS	ASSEMBLY PROCESS	SPEC	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
				MEC	GL/QA	
1	All parts	- All parts must be free from dust, rust and dent - All parts have been cleaning - No crack, no scratch, no wear - Always check the contact of the following parts before assembly.				
2	Tension Lever	- Make sure tension lever is clean and free from damage. - Reamer bearing seat hole, then install outer race and needle bearing (7) - *Using press tool to install bearing. - Install dust, seal (8) before assembly with shaft				
3	Pulley ass'y	- Make sure tension lever is clean and free from damage. - Install snap ring (15) on groove in the pulley - Install bearing (12) and spacer (13) in the pulley - *Using press tool to install bearing. - Install Oil seal (14) before assembly with lever				
4	Assembly Pulley and Lever	- Assembly pulley ass'y (11) and tension lever (2) - Install cover lock (16), then tighten mounting bolts (17). Actual : kgm	3.2 ± 0.3 kgm			
5	Pulley Cover and Relief Valve	- Fit gasket (20), then install cover (18) - Install mounting bolts (21), then tighten Actual : kgm - Install relief valve (19) then tighten	6.7 ± 0.7 kgm			
6	Assembly Lever assy and tension shaft	- Install Inner race (5) to the shaft (1) - Install spring (5), then assembly lever ass'y (2) - Install plate (9), install mounting bolts (10) then tighten. Actual : kgm	6.7 ± 0.7 kgm			

ASSEMBLY CHECK SHEET TENSION PULLEY ENGINE 6D140-2 SERIES	WO NUMBER		DOCUMENT NO	
	ENGINE MODEL		PUBLISHED DATE	20/10/2009
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	2 of 2

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
				MEC	GL/QA	
7	Fitting grease and plug	- Install fitting grease and plug, then tighten - Give all of surface pulley with grease to protect from rust				

Notes :

ASSEMBLY BY	INSPECTED & APPROVED BY	
_____ MECHANIC (MEC)	_____ QUALITY ASSURANCE (QA)	_____ PROD. SUPERVISOR (SPV)

REV. 00



PLAN REBUILD CENTER

FAN PULLEY

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

FAN PULLEY ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

JOB NUMBER		PART NAME	FAN PULLEY	DOCUMENT NO	
ENGINE MODEL	6D140-2	PART NUMBER		PUBLIC DATE	08/09/2009
ENGINE SERIAL NO.		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

ASSEMBLY BY	All dimension in millimetres
	All Torques in Kilogram metres

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC	SIGN			REMARKS (SKETCH, PHOTOS, COMMENT)									
					MEC	QA	SPV										
1	ALL PARTS		- Make sure all parts are ready to used (free Crack ; worn; corrosion and dirt) - Specification of spare parts are correct - Before assembling , wash and clean the parts. - Allow them to dry by air blow														
2	Case Housing		- Housing condition - No crack, no scratch, no corrosion. - Check all holes and then blow with air pressure														
3	Shaft and Boss		- Check contact surface, make sure that clean and free corrosion and damage. - Measure dimension shaft and boss <table border="1"> <tr> <td>O.D Shaft</td> <td>Actual :</td> <td>mm</td> </tr> <tr> <td>I.D Boss</td> <td>Actual :</td> <td>mm</td> </tr> <tr> <td>Interference</td> <td>Actual :</td> <td>mm</td> </tr> </table>	O.D Shaft	Actual :	mm	I.D Boss	Actual :	mm	Interference	Actual :	mm	O.D Shaft : 45.018 ~ 45.002 mm I.D Boss : 44.952 ~ 44.977mm Interference : 0.025 ~ 0.066 mm				
O.D Shaft	Actual :	mm															
I.D Boss	Actual :	mm															
Interference	Actual :	mm															
4	Fan Pulley drive assembly		- Check all holes and then blow with air pressure - Install bearing (4) to case housing using special tool . - Install shaft (2) and bearing (5) to case housing using special tool . - Install plate (3) to shaft and fit it with housing - Install mounting bolts (7) and tighten - Torque bolt : <table border="1"> <tr> <td>Actual :</td> <td>kgm</td> </tr> </table> - Finally Install boss to shaft and case housing using special tool .	Actual :	kgm	2.8 ~ 3.5 Kgm											
Actual :	kgm																
5	Greasing puley		- Give all of surface pulley with grease to protect from rust														

MECHANIC ASSEMBLY COMMENT	

ASSEMBLY BY _____ MECHANIC (MECH)	INSPECTED & APPROVED BY _____ QUALITY ASSURANCE (QA) PROD. SUPERVISOR (SPV)
--	--

REV. 00



PLAN REBUILD CENTER

AFTERCOOLER

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

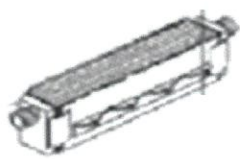
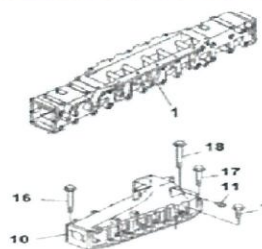
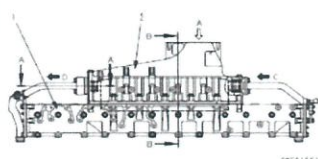
WO NO:

SUB ASS'Y SECTION

ASSEMBLY CHECK SHEET
AFTER COOLER
KOMATSU ENGINE 6D140E-3

JOB NUMBER		PART NAME		DOCUMENT NO	RC/QR04/EG/18/03
ENGINE MODEL	SA6D140-2	PART NUMBER		PUBLISHED DATE	01/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL	GD 825A-2	FINISH DATE / TIME		PAGE	2 of 2

PROCESS	START	ASSEMBLY BY
	FINISH	

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOES,
					MEC	QA	SPV	
1	Core Element		- Test and check flow in core element. Standard water temperaure = 65 ~ 75°C - Test and check leakage in core element Standard pressure test = 3 ~ 4 kg/cm² Act. : Kg/cm² - Check o ring seat - Check damage and crack on core, tube, fin and bracket.	After cooler tester				 
2	Intake Housing		- Check all threads on intake housing - Check cleanliness, damage and crack on intake housing.					
3	Intake Cover		- Check all threads on intake core - Check cleanliness damage, and crack on intake cover.					
4	After Coller Assembly		- Install core element - Install bolt and tighten Standard torque Bolt M17 side 6 = 4.5± 0.5 Kgm Act. : Kgm - Install connector on housing					
5	Plug		- Install plug on housing and tighten. Standard torque Plug NPT 5 mm = 1 ± 0.5 Kgm Act. : Kgm					

JOB NUMBER		PART NAME		DOCUMENT NO	RC/QR04/EG/18/03
ENGINE MODEL	SA6D140-2	PART NUMBER		PUBLISHED DATE	01/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL	GD 825A-2	FINISH DATE / TIME		PAGE	2 of 2

[illegible]

ASSEMBLY BY		INSPECTED BY & APPROVED BY	
			
MECHANIC		QUALITY ASSURANCE (QA)	
			
		GL/SPV PRODUCTION	

REV. 00



PLAN REBUILD CENTER

OIL FILTER HEAD

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

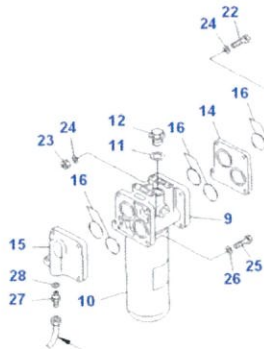
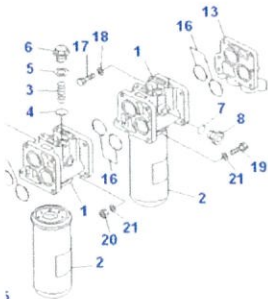
SUB ASS'Y SECTION

ASSEMBLY CHECK SHEET

OIL FILTER HEAD OF ENGINE 6D140-2

WO NUMBER		PART NAME	OIL FILTER HEAD	DOCUMENT NO.	RC/QR04/EG/14/05
ENGINE MODEL	6D140-2	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

ASSEMBLY BY	
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NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	SPV / QA	
1	Brackets		- Confirm that bracket is free from dirt, crack, corrosion and damage. - Confirm that all threads are free from damage.				
2.	Oil Filter Head Assembly	Connector	- Tightening connector to bracket Actual : Kgm - Tightening cap to Oil Filter Head Actual : Kgm - Tightening Oil Filter Head mounting bolt Actual : Kgm - Tightening plug 14 mm to Oil Filter Head Actual : Kgm	9 ± 0.5 Kgm 5 ± 0.5 Kgm 5 ~6.5 Kgm 3 ± 0.5 Kgm			
3.	Final Inspection		- Recheck all plugs and cap from loosen - Recheck marks				

[illegible]

ASSEMBLY BY

INSPECTED & APPROVED BY :

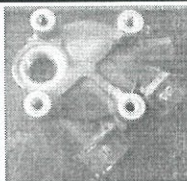
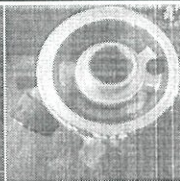
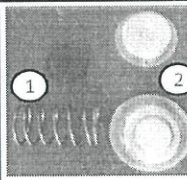
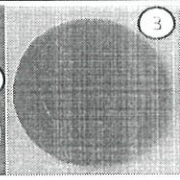
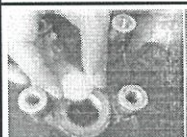
MECHANIC (MECH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

ASSEMBLY CHECK SHEET OIL FILTER HEAD KOMATSU ENGINE 6D140 SERIES	WO NUMBER		Eks UNIT CODE	
	ENGINE MODEL	6D140 SERIES	PUBLISHED DATE	23-Mar-16
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		DOCUMENT NO	

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	MAIN HOUSING		- Check main housing condition from rust and dirty. - Make sure O ring area free from crack, pitted and corrosion. - Check all thread condition from crack and broken.	Visual check				 
2	Spring, Valve and plug.		- Check spring (1) from rust and broken. make sure spring no bend and no broken. - Check big and small plug from rust, make sure thread good condition. - Check valve from corrosion, and make sure rubber still good condition.	Visual check				 
3	Assembly		- Coat valve with engine oil and install to housing # valve with rubber side on lower position. - Coat spring with engine oil and install on valve install O ring on seat then install big plug. Mounting torque plug: Actual : Kgm - Install O-ring on small plug and install on small thread on main housing. Mounting torque plug: Actual : Kgm	Visual check 3.5 ~ 5 kgm (Reman manual)				    
4	Final Inspection		After finish assembly the oil filter head make sure all surface on main housing was coating with engine oil for prevent corrosion and use white tag (RFA TAG) on this oil filter head.	Visual check				

Comment :

ASSEMBLY BY	INSPECTION & APPROVED BY
MECHANIC (MEC)	QUALITY ASSURANCE (QA) PROD. SUPERVISOR (SPV)

REV. 00



PLAN REBUILD CENTER

OIL ADAPTER

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D140-2

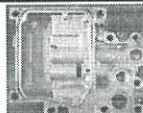
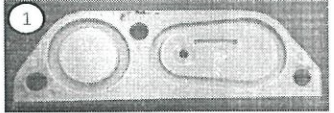
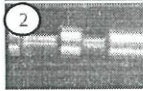
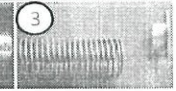
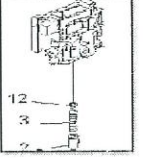
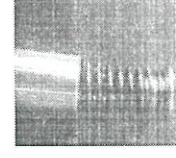
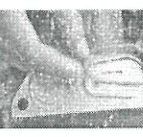
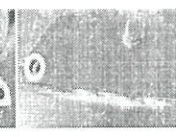
MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

ASSEMBLY CHECK SHEET OIL ADAPTER KOMATSU ENGINE 6D140 SERIES	WO NUMBER		Eks UNIT CODE	
	ENGINE MODEL	6D140 SERIES	PUBLISHED DATE	23-Mar-16
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		DOCUMENT NO	

PROCESS	START FINISH		ASSEMBLY BY	
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NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	MAIN HOUSING		- Check main housing condition from rust and dirty. - Make sure O ring area free from crack, pitted and corrosion.	Visual check				 
2	Cover, Valve, Spring and spring seat		- Check cover (1) from rust and broken. make sure O ring seat good condition. - Check valve (2) good condition, no scratch and no corrosion. - Check spring and seat spring condition. make sure spring no bend and none corrosion on spring seat.	Visual check				  
3	Assembly		- Coat spring with engine oil and install to valve, then install spring seat on spring end. - install valve and spring to main housing make sure spring seat in right position. <i># Push spring tension for recheck the spring not jammed.</i> - Install Large and small O ring to cover. - Install cover and mounting bolt to main housing - Setting mounting bolt torque Actual : Kgm	Visual check 3 ~ 3.5 kgm (Reman manual)				    
4	Final Inspection		After finish assembly the oil adapter make sure all surface on main housing was coating with engine oil for prevent corrosion and use white tag (RFA TAG) on this oil adapter.	Visual check				

Comment :

ASSEMBLY BY	INSPECTION & APPROVED BY
MECHANIC (MEC)	QUALITY ASSURANCE (QA) PROD. SUPERVISOR (SPV)

RC/QA/EG/050

REV. 01



PT.SAPTAINDRA SEJATI

CROSS HEAD SALVAGING PROCESS AND INSPECTION CHECK SHEET KOMATSU ENGINE ALL SERIES

MACHINE MODEL

WO :

SUB ASSY SECTION

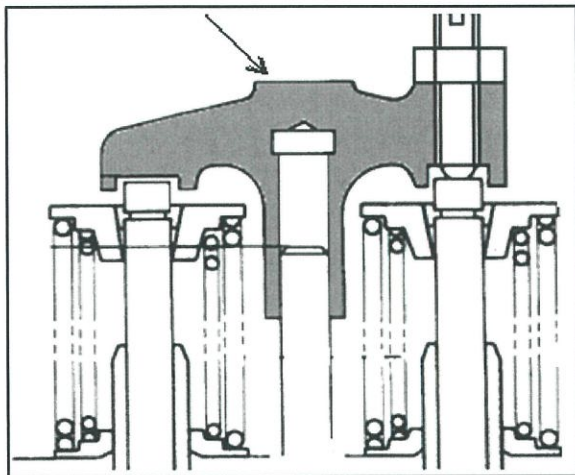


PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

SALVAGE CROSS HEAD PROCESS INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CROSS HEAD	DOCUMENT NO.	QR06-SAL-006
ENGINE MODEL		PART NUMBER		PUBL. DATE	01/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

SKETCH DRAWING



Standard

Upper pad portion : Free from Wear

MECHANIC NAME

VISUAL CHECK	MECHANIC		QA	
	OK	NOT OK	OK	NOT OK
CRACK HAS BEEN CHECKED				
CHECK THREAD OF CROSSHEAD				
CHECK THREAD OF SCREW				
CHECK THREAD OF NUT				

MAN HOURS UTILIZATION

PROCESS	START	FINISH	MAN HOURS
Polishing			
Measuring			
TOTAL			

Physical Data

No.	Part Number	Part Name	Qty	New	Reuse
1		Cross head			
2		Screw			
3		Nut			

MECHANIC INSPECTION COMMENT :

INSPECTION BY

INSPECTION & APPROVED BY

MECHANIC

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV. 00



PLAN REBUILD CENTER

INJECTION PRESSURE NOZZLE

TEST REPORT CHECK SHEET

KOMATSU ENGINE 6D140-2

MACHINE MODEL

WO NO:

SUB ASS'Y SECTION

TEST REPORT CHECK SHEET INJECTION PRESSURE NOZZLE KOMATSU ENGINE 6D140-2	DOCUMENT NO.	RC/QR05/EG/03/02
	PUBLICATION DATE	01/04/2009
	REVISION NO.	00
	PAGE	1 of 1

WO NUMBER		PART NAME	
ENGINE MODEL		PART NO.	
ENGINE S/N		START DATE & TIME	
MACHINE MODEL		FINISH DATE & TIME	

TESTED BY		QUANTITY	

INJECTION PRESSURE TEST

NO.	STANDARD VALUE	CHECK ITEM	NOZZLE / INJECTOR NO.					
			1	2	3	4	5	6
	6D140-2, S6D140-2, SA6D140-2, SAA6D140-2 Cracking Pressure : (255 ~ 263 kg/cm ²) (24.55 ~ 25.80 Mpa)	INJECTION PRESSURE NOZZLE						
		START LEAKING PRESSURE						
		CONDITION OF SPRAY						

ADJUSTMENT SHIM

NO.	STANDARD VALUE	CHECK ITEM	NOZZLE / INJECTOR NO.					
			1	2	3	4	5	6
	SHIM THICKNESS OF 0.025 mm = 3.5Kg/cm ² (3.4 BAR)	THICKNESS OF SHIM						

COMMENT :

TESTED BY	INSPECTED & APPROVED BY	
MECHANIC (MECH)	QUALITY ASSURANCE (QA)	GL/SPV PRODUCTION